

BHUMI-FUSE

AI-Driven Multi-Source Urban Land Record Harmonization Engine SMART INDIA HACKATHON 2026 • PS-26013 • COMPLETE BEGINNER & JUDGE GUIDE

ONE-LINE PITCH: BHUMI-FUSE converts conflicting cadastral maps, drone orthomosaics, GNSS survey points, and municipal GIS layers into an auditable, topology-safe, harmonized land representation — without ever overwriting an authoritative legal record.

1. The Core Problem: Two Spatial Realities

In India, every parcel of land exists in **two separate spatial realities**:

- **Legal / Recorded Reality:** The historical government cadastral map (e.g. 1960 sheet, Village Map, 7/12 extract) that legally defines ownership boundaries.
- **Physical Reality:** What actually exists on the ground today, captured by modern drone orthomosaics, satellite imagery, and high-precision GNSS surveys.

Because these sources were captured decades apart using different tools, accuracies, coordinate systems, and scales, they **rarely align perfectly**. Traditional GIS tools (like ArcGIS or QGIS) simply overlay these maps, but they cannot decide *how* to reconcile the conflicts safely. BHUMI-FUSE builds an intelligent geospatial decision-support pipeline that analyzes cross-source evidence, computes safe registration, protects topology, and flags ambiguous cases for authorized human review.

2. The 8-Stage Harmonization Workflow

Stage	Component	How It Works & What It Does	Safety / Trust Guarantee
1. Ingest	Data Fabric	Accepts multi-format data: Cadastral (.shp/.kml), Drone (.tif), GNSS (.csv), Municipal (.gpkg).	Immutable write-once tables; originals are never overwritten.
2. Normalize	CRS Transformer	Detects source coordinate systems (e.g. UTM Zone 43N EPSG:32643) and transforms all layers into unified WGS84 (EPSG:4326).	Eliminates 'fake' coordinate mismatches before comparing geometry.
3. Extract	GeoAI Perception	Runs lightweight pretrained vision models (SegFormer-B0 / SAM2) on drone imagery to extract physical boundary contours with confidence scores.	AI proposals are treated as evidence, NOT authoritative decisions.
4. Graph	Spatial Evidence Graph	Connects parcels, AI contours, GNSS control points, and roads into a unified NetworkX relationship graph.	Replaces isolated map layers with connected multi-source context.
5. Match & Register	Registration Engine	Uses spatial indexing and RANSAC Thin Plate Spline (TPS) elastic deformation to align historical records with ground control.	Computes quantitative RMSE and residual error vectors.
6. Validate	Topology Guard	Executes deterministic PostGIS / GEOS topology checks (ST_IsValid, ST_Overlaps, gap detection, adjacency preservation).	Rejects any alignment that introduces overlaps, slivers, or broken rings.
7. Fuse & Score	Evidence Fusion	Computes a composite trust score based on authority, accuracy, age, and cross-source agreement. Cases below 62% trigger 'Do Not Decide'.	Exposes uncertainty transparently instead of forcing a guess.
8. Review & Export	Governance Layer	Authorized officer reviews evidence cards, chooses Accept / Reject / Adjust / Escalate, and exports ISO-compliant GeoJSON / GeoPackage.	Creates an immutable, versioned audit trail for every action.

3. Detailed Screen-by-Screen Architecture

- **Screen 0 — Dashboard (Overview):** High-level summary displaying 4 KPI cards (24 Active Investigations, 12,845 Parcels Processed, 128 High Priority Cases, 68% Auto-Resolved), an interactive Discrepancy Heatmap showing mismatch intensity across Kharadi Sector 12, an Investigation Status Donut Chart, and Recent Investigations table.
- **Screen 1 — Upload & Ingest Sources:** A 3-stage upload wizard for Cadastral Shapefiles (1960), Drone Orthomosaics (2024), GNSS CSV points, and Municipal GeoPackages. Displays detected CRS, total file size (2.48 GB), and triggers automated CRS normalization.
- **Screen 2 — Source Viewer:** Interactive side-by-side and 4-pane comparative viewer. Allows toggling individual layers with real-time opacity sliders (Cadastral 60%, Drone 70%, Municipal 50%, GNSS Points) to visibly highlight historical distortion before registration.
- **Screen 3 — AI Boundary Extraction:** Displays SegFormer-B0 deep learning boundary segmentation pipeline. Features an Extraction Status checklist (128 features, 64 boundaries, 89% avg confidence) alongside high-contrast dark-mode glowing vector contour previews and confidence heatmaps.
- **Screen 4 — Spatial Evidence Graph:** Interactive D3/Force-directed relational network. Visualizes parcels, AI boundaries, GNSS points, and roads as connected nodes with relationship edges (matches, supports, intersects, adjacent_to). Includes a live Node Property Inspector.
- **Screen 5 — Harmonization & Alignment:** 5-step sub-tab interface (Matching, Registration, Topology Check, Fusion, Scoring). Features an alignment quality panel (RMSE: 0.82 m, Inlier Ratio: 92%, Max Residual: 3.67 m), residual distribution histogram, and side-by-side Before vs After alignment comparison.
- **Screen 6 — Legal vs Physical Discrepancy Map:** Comprehensive government discrepancy map. Classifies parcels into High Conflict (>3m Red), Medium Conflict (1-3m Orange), Low Conflict (<1m Yellow/Green), and No Conflict (Green). Hovering or clicking any parcel opens a real-time displacement intelligence popup.
- **Screen 7 — Evidence Card & Recommendation:** Detailed investigation card for a single parcel (e.g. Parcel 101). Displays 5-star metric ratings for Authority, GNSS Support, AI Confidence, Mean Displacement (2.45m), Topology Status (Pass), and an overall confidence gauge (34% - Review Required).
- **Screen 8 — Review & Decision:** Dedicated authorized officer adjudication terminal. Features a zoomed multi-layer map overlay (dashed red legal, solid blue drone, solid green harmonized) with interactive decision options: Accept, Reject, Adjust, Escalate, and Do Not Decide, with reviewer notes.
- **Screen 9 — Audit Trail & Provenance:** Immutable cryptographic chronological log. Records every ingest, transformation, alignment parameter, topology validation, and human review decision with timestamp, user ID, and action details. Supports one-click GeoJSON and GeoPackage export.

4. The 10 Unique Innovations (Why BHUMI-FUSE is NOT Just GIS)

Innovation	What It Does & Why It Wins 20 Marks for Innovation
1. Spatial Evidence Graph	Treats parcels, boundaries, survey points, and roads as connected graph nodes rather than isolated flat map layers, enabling multi-hop spatial reasoning.
2. Authority-Aware Fusion	Does not blindly trust newer drone data over legal cadastral records. Balances source authority, positional accuracy, age, and cross-source consensus.
3. Topology-Constrained Harmonization	Alignment is accepted ONLY when resulting geometries pass strict PostGIS/GEOS validity rules without creating overlaps, gaps, or destroying neighbor adjacency.
4. Boundary-Level Uncertainty	Attaches uncertainty scores to individual boundary segments, exposing exactly which edge is trustworthy and which edge is doubtful.
5. Temporal Conflict Engine	Differentiates between mapping/registration errors and legitimate physical development over time (e.g. new construction built in 2024).
6. Legal-vs-Physical Discrepancy Map	Turns spatial disagreements into a first-class, color-coded intelligence layer that immediately highlights municipal conflict hotspots.
7. Risk-Based Human Review	Prioritizes cases using an intelligent formula: Impact × Uncertainty × Displacement × Legal Sensitivity, so officers review highest-risk cases first.
8. 'Do Not Decide' Mode	When evidence is ambiguous or confidence is below threshold (<62%), the system explicitly defers to human officers rather than hallucinating an answer.
9. Explainable Evidence Cards	Replaces black-box AI scores with a transparent breakdown of authority, GNSS support, CV confidence, displacement, and topology checks.
10. Immutable Provenance & Versioning	Original evidence is never modified or deleted. Every algorithmic transformation and officer review creates an append-only, auditable version.

5. Data Strategy: What is Real vs Simulated

In compliance with SIH evaluation rules, every dataset layer in the Pune / Kharadi pilot is clearly categorized by provenance:

- **Real OpenStreetMap Building Footprints:** Authentic vector building polygons extracted directly from OpenStreetMap via Overpass API for Kharadi Sector 12.
- **Real Municipal Road Network:** Actual road centerlines and highway corridors (Kharadi Bypass, Mundhwa Road).
- **Real Satellite Imagery Basemap:** Live high-resolution Esri World Imagery tiles (0.1m GSD equivalent) streamed via MapLibre GL.
- **Simulated Historical Cadastral Baseline (1960):** Synthetically shifted and rotated parcel polygons derived from real footprints to represent realistic historical land record distortions (clearly labeled simulated to maintain scientific integrity).
- **Simulated GNSS Precision Control:** Real landmark coordinates augmented with synthetic RTK receiver precision metadata (0.04m accuracy).

6. SIH 2026 Judge Scorecard Alignment (100 Marks)

Evaluation Criteria (20 Marks Each)	How BHUMI-FUSE Secures 18–20 Marks ('Excellent' Band)
1. Problem Understanding & Relevance (20)	Explicitly addresses the dual-reality problem (Legal vs Physical), targets municipal land authorities (PMC/PMRDA), and directly aligns with national initiatives (DILRMP & SVAMITVA).
2. Innovation & Uniqueness (20)	Presents 10 clear novel features: Evidence Graph, Authority-Aware Fusion, Topology Guard, Do-Not-Decide mode, and Explainable Evidence Cards.
3. Feasibility & Technical Approach (20)	100% free open-source stack (React, TypeScript, Vite, MapLibre GL, Python, FastAPI, Shapely, GeoPandas, NetworkX). No paid API tokens or giant unreachable GPU training required.
4. Impact & Scalability (20)	Drastically cuts officer reconciliation time, prevents boundary disputes before they reach court, and provides a scalable architecture from single ward to district/state scale.
5. Solution Clarity & Execution (20)	Seamless 10-screen live working demo with interactive maps, before/after alignment toggle, decision submission, and downloadable audit trail.

7. The 30-Second Winning Pitch to Judges

Judge Pitch Script:

'Respected Judges, India has two spatial realities: the legal land record and the physical ground reality. They were created at different times, scales, and coordinate systems, and they constantly disagree. Existing GIS tools can overlay them, but neither decides how to reconcile them safely. BHUMI-FUSE builds a Spatial Evidence Graph connecting cadastral vectors, drone imagery, GNSS points, and municipal GIS. We normalize CRS, extract physical boundaries with AI, compute robust topology-safe registration, combine evidence by legal authority and confidence, and route uncertain cases to authorized officers. We never claim AI decides ownership; we empower officials with an auditable, versioned decision-support workflow.'